

# PART IIB REVISION NOTES

## 4M24: Computational Statistics and Machine Learning

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### 1 Monte Carlo Methods

**Central limit theorem for Monte Carlo:** The Monte Carlo estimate follows

$$\frac{1}{N} \sum_n f(x_n) \sim \mathcal{N}\left(\mathbb{E}[f(x)], \frac{\sigma_f^2}{N}\right) \text{ as } N \rightarrow \infty$$

where  $\sigma_f^2$  is the variance of  $f(x)$ .

*\*Proof:* Define a transform to zero mean and variance

$$\tilde{f}(x) = \frac{f(x) - \mathbb{E}[f(x)]}{\sigma_f}$$

Define  $S_N = \sum_{n=1}^N \tilde{f}(x_n)$  and  $Z_N = \frac{S_N}{\sqrt{N}}$ . The moment generating function of  $Z_N$  is

$$\begin{aligned} M_{Z_N}(t) &= \mathbb{E}[\exp(tZ_N)] = \mathbb{E}\left[\exp\left(t \frac{S_N}{\sqrt{N}}\right)\right] \\ &= \mathbb{E}\left[\exp\left(\frac{t}{\sqrt{N}} \sum_{n=1}^N \tilde{f}(x_n)\right)\right] \\ &= \prod_{n=1}^N \mathbb{E}\left[\exp\left(\frac{t}{\sqrt{N}} \tilde{f}(x_n)\right)\right] = \left(\mathbb{E}\left[\exp\left(\frac{t}{\sqrt{N}} \tilde{f}(x)\right)\right]\right)^N \end{aligned}$$

Taylor expansion of the exponential function gives

$$\exp\left(\frac{t}{\sqrt{N}} \tilde{f}(x)\right) = 1 + \frac{t}{\sqrt{N}} \tilde{f}(x) + \frac{t^2}{2N} \tilde{f}(x)^2 + O(N^{-2})$$

Since  $\mathbb{E}[\tilde{f}(x)] = 0$  and  $\mathbb{E}[\tilde{f}(x)^2] = 1$ , the expectation is

$$\mathbb{E}\left[\exp\left(\frac{t}{\sqrt{N}} \tilde{f}(x)\right)\right] = 1 + \frac{t^2}{2N} + O(N^{-2})$$

Hence, the MGF of  $Z_N$  is

$$M_{Z_N}(t) = \left(1 + \frac{t^2}{2N} + O(N^{-2})\right)^N \rightarrow \exp\left(\frac{t^2}{2}\right) \text{ as } N \rightarrow \infty$$

which is the MGF of the standard normal. Hence  $Z_N \rightarrow \mathcal{N}(0, 1)$ ,  $S_N \rightarrow \mathcal{N}(0, N)$ ,  $\frac{1}{N} \sum_{n=1}^N f(x_n) \rightarrow \mathcal{N}(\mathbb{E}[f(x)], \frac{\sigma_f^2}{N})$ .

**Importance sampling:** use another probability density  $q(x)$  that satisfy  $q(x) \neq 0$  when  $f(x) \neq 0$  to sample  $x$ . The expectation can be rewritten as

$$\begin{aligned}\mathbb{E}_p[f(x)] &= \int f(x)p(x)dx \\ &= \int f(x)\frac{p(x)}{q(x)}q(x)dx \\ &= \mathbb{E}_q\left[f(x)\frac{p(x)}{q(x)}\right]\end{aligned}$$

The original Monte Carlo estimate  $\frac{1}{N} \sum_n f(x_n)$  with each  $x_n \sim p(x)$  becomes  $\frac{1}{N} \sum_n f(x_n) \frac{p(x_n)}{q(x_n)}$  with each  $x_n \sim q(x)$ . The difference in variances is

$$\sigma_f^2 - \sigma_{\hat{f}}^2 = \int f(x)^2 p(x)dx - \int f(x)^2 \frac{p(x)^2}{q(x)}dx$$

The biggest decrease in variance is achieved when  $q(x) \propto |f(x)|p(x)$ , leading to  $\sigma_{\hat{f}}^2 = 0$ .

**Control variates:** use another function  $g(x)$  with known expectation  $\mathbb{E}[g(x)]$  to reduce the variance of the Monte Carlo estimate. The new estimator is

$$\hat{E}_f^c = \hat{E}_f + c(\hat{E}_g - \mathbb{E}[g(x)])$$

where  $\hat{E}_f$  and  $\hat{E}_g$  are the Monte Carlo estimates of  $f(x)$  and  $g(x)$ , respectively.

The optimal value of  $c$  is

$$c^* = -\frac{\text{Cov}(\hat{E}_f, \hat{E}_g)}{\text{Var}(\hat{E}_g)}$$

giving the optimal estimator variance of

$$\text{Var}(\hat{E}_f^{c^*}) = \text{Var}(\hat{E}_f) - \frac{\text{Cov}(\hat{E}_f, \hat{E}_g)^2}{\text{Var}(\hat{E}_g)}$$

For multiple control variates, the estimator is  $\hat{E}_f^c = \hat{E}_f + \sum_i c_i(\hat{E}_{g_i} - \mathbb{E}[g_i(x)])$ , and the optimal coefficients are  $\mathbf{c}^* = -C^{-1}\mathbf{b}$ , where  $C$  is the covariance matrix of  $\hat{E}_{g_i}$  and  $\mathbf{b}$  is the covariance vector between  $\hat{E}_f$  and  $\hat{E}_{g_i}$ . The optimal estimator variance is  $\text{Var}(\hat{E}_f^{c^*}) = \text{Var}(\hat{E}_f) - \mathbf{b}^\top C^{-1}\mathbf{b}$ .

## 2 Lebesgue Integral and Measure

**Riemann integral:** Define two summations for a function  $f$  on an interval  $[a, b]$  and  $\Delta x_k = [x_k, x_{k+1}]$ :

$$\begin{aligned}S_p(x) &= \sum_{k=1}^n M_k(x_{k+1} - x_k) \text{ with } M_k = \sup_{x \in \Delta x_k} f(x) \\ s_p(x) &= \sum_{k=1}^n m_k(x_{k+1} - x_k) \text{ with } m_k = \inf_{x \in \Delta x_k} f(x)\end{aligned}$$

Define the limit of the summations as  $n \rightarrow \infty$ :  $S(f) = \liminf_{n \rightarrow \infty} S_p(x)$ ,  $s(f) = \limsup_{n \rightarrow \infty} s_p(x)$ . If  $S(f) = s(f) = A$ , then  $A$  is the Riemann integral  $\int_a^b f(x)dx$ .

Limitations:

- The function must be continuous almost everywhere.
- The construction of the integral requires a partition of the domain.
- The exchange of limit and integral is not always valid.

**Convergence of functions:**

- *Pointwise convergence*: Given  $\epsilon > 0$ , there exists a natural number  $N(\epsilon, x)$  such that for  $n > N$ ,  $|f_n(x) - f(x)| < \epsilon$ . Does not preserve continuity.
- *Uniform convergence*: Given  $\epsilon > 0$ , there exists a natural number  $N(\epsilon)$  such that for  $n > N$ ,  $|f_n - f| < \epsilon, \forall x \in D$ . Preserves continuity and *allows exchange of limit and integral*.

**Lebesgue integral:** Partition the range of the function instead of the domain. For  $0 \leq f_{\min} \leq f(x) \leq f_{\max}$ , the range is partitioned into  $f_{\min} = f_1, f_2, \dots, f_n = f_{\max}$ . For each  $f_k$ , there are sets of values  $X_k$  such that  $f_k \leq f(x) < f_{k+1}$  for  $x \in X_k$ , and the size (measure) of each set is  $\mu(X_k)$ .

The Lebesgue integral is the limiting value of the sum:

$$\int f d\mu = \lim_{n \rightarrow \infty} \sum_{k=1}^n f_k \mu(X_k) = \lim_{n \rightarrow \infty} \sum_{\max |f_k - f_{k-1}| \rightarrow 0}^n f_k \mu(X_k)$$

Advantages:

- Only requires a measure of sets on the domain of integration – more general
- Change of order of limit and integration only requires pointwise convergence.
- A Riemann integrable function is also Lebesgue integrable, but not vice versa.

**Measure of sets:** A set  $X$  and a set  $\Sigma$  generated from  $X$  (contains  $X, \emptyset$ , and other subsets of  $X$ , closed under countable complementation, union and intersections) defines a *measurable set*  $\{X, \Sigma\}$ . Along with a measure  $\mu$  it defines a *measure space*  $\{X, \Sigma, \mu\}$ .

Rules for measure:

- If  $\mu(E) \geq 0$  for all  $E \in \Sigma$ , all elements in  $\Sigma$  are measurable.
- $\mu(\emptyset) = 0$ .
- If  $E_i \subseteq E_j$ , then  $\mu(E_i) \leq \mu(E_j)$ . (Monotonicity)
- $\mu(\bigcup_{i=1}^{\infty} E_i) = \sum_{i=1}^{\infty} \mu(E_i)$  for disjoint sets  $E_i \in \Sigma$ . (Countable additivity)
- If  $(X, \Sigma_X)$  and  $(Y, \Sigma_Y)$  are both measurable, then a function  $f : X \rightarrow Y$  is measurable if  $f^{-1}(B) \in \Sigma_X$  for all  $B \in \Sigma_Y$ .

**Lebesgue measure:**

- For a continuous line segment on  $\mathbb{R}$ , the Lebesgue measure is the length of the segment. e.g.  $I = [a, b]$ ,  $\mu(I) = b - a$ .

- The Lebesgue measure of a single point is zero. By additivity, the Lebesgue measure for the set of rational numbers  $\mu(\mathbb{Q}) = 0$ .
- The Lebesgue measure of countable sets (any set that can be put into a one-to-one correspondence with the set of integers  $\mathbb{Z}$ ) is zero.

**Probability measures:** The *probability space*  $\{\Omega, \mathcal{F}, P\}$  consists of three elements:

- Sample space  $\Omega$ : the set of all possible outcomes.
- $\sigma$ -algebra  $\mathcal{F}$ : certain subsets of  $\Omega$  with outcomes to be considered.
- Probability measure  $P$ : a function that assigns a probability to each event in  $\mathcal{F}$ , satisfying  $P(\Omega) = 1$  and countable additivity.

**Random variables:** For a probability space  $\{\Omega, \mathcal{F}, P\}$  and a measurable space  $\{E, \mathcal{E}\}$ , there is an associated measurable function  $X : \Omega \rightarrow E$  that satisfies  $X^{-1}(B) \in \mathcal{F}$  for all  $B \in \mathcal{E}$ . If  $E = \mathbb{R}$ , then  $X : \Omega \rightarrow \mathbb{R}$  is a real-valued *random variable* such that  $\{\omega \in \Omega : X(\omega) \leq r\} \in \mathcal{F}$  for every  $r \in \mathbb{R}$ .

The expectation of a random variable  $X$  is defined as  $\mathbb{E}[X] = \int_{\Omega} X(\omega) dP(\omega)$ .

**Radon-Nikodym theorem:** Consider two measures  $\mu$  and  $\nu$  on a measurable space  $\{X, \Sigma\}$ . If for every  $A \in \Sigma$  where  $\nu(A) = 0$  also have  $\mu(A) = 0$ , then  $\mu$  is *absolutely continuous* with respect to  $\nu$ , denoted as  $\mu \ll \nu$ .

Given a probability measure  $P$ , a new probability measure  $Q$  can be defined if there exists a function  $L(\omega)$  such that  $dQ(\omega) = L(\omega)dP(\omega)$ .

For any measurable set  $B$ ,  $Q(B) = \int_B L(\omega)dP(\omega)$  is the probability of  $B$  under the new measure  $Q$ .

The expectation of a function  $F(\omega)$  can be computed as

$$\int_{\Omega} F(\omega) dQ(\omega) = \int_{\Omega} F(\omega) L(\omega) dP(\omega)$$

where the function  $L(\omega) = \frac{dQ(\omega)}{dP(\omega)}$  is the Radon-Nikodym derivative if  $Q \ll P$ , subject to (1)  $L(\omega) \geq 0$  for all  $\omega \in \Omega$  and (2)  $\int_{\Omega} L(\omega) dP(\omega) = 1$ .

For a random variable  $X$  on  $\mathbb{R}$ , the probability of a small neighborhood  $B$  around the outcome  $X = x \in \mathbb{R}$  is  $\int_B dP = \int_B L(x) d\mu(x) = \int_B L(x) dx$ . The function  $L(x) = \frac{dP(x)}{dx}$  is the probability density function of  $P$  with respect to the Lebesgue measure.

### 3 Markov Chain Monte Carlo

**Probability inversion:** For data  $\mathbf{x} \in \mathcal{X}$  and parameters  $\boldsymbol{\theta} \in \Theta$ , the posterior distribution is  $\pi(\boldsymbol{\theta}|\mathbf{x}) = \frac{p(\mathbf{x}|\boldsymbol{\theta})\pi_0(\boldsymbol{\theta})}{\int_{\Theta} p(\mathbf{x}|\boldsymbol{\theta})\pi_0(\boldsymbol{\theta})d\boldsymbol{\theta}}$ . Inference requires integration with respect to the posterior

$$I(\boldsymbol{\theta}) = \mathbb{E}_{\boldsymbol{\theta}|\mathbf{x}}[f(\boldsymbol{\theta})] = \int_{\Theta} f(\boldsymbol{\theta})\pi(\boldsymbol{\theta}|\mathbf{x})d\boldsymbol{\theta}$$

The Monte-Carlo estimate is

$$\hat{I}_N(\boldsymbol{\theta}) = \frac{1}{N} \sum_{n=1}^N f(\boldsymbol{\theta}_n) \text{ with } \boldsymbol{\theta}_n \sim \pi(\boldsymbol{\theta}|\mathbf{x})$$

The estimator is unbiased and consistent as  $N \rightarrow \infty$ ,  $\hat{I}_N(\boldsymbol{\theta}) \rightarrow I(\boldsymbol{\theta})$  and the error  $\sqrt{N}(\hat{I}_N(\boldsymbol{\theta}) - I(\boldsymbol{\theta})) \rightarrow \mathcal{N}(0, \sigma_f^2)$ .

**Markov chains:** stochastic transitions governed by the Markov property

$$p(\boldsymbol{\theta}^{(i)} | \boldsymbol{\theta}^{(i-1)}, \dots, \boldsymbol{\theta}^{(1)}) = T(\boldsymbol{\theta}^{(i)} | \boldsymbol{\theta}^{(i-1)})$$

*Invariant distribution:* the Markov chain converges to the invariant distribution  $\pi(\boldsymbol{\theta}^{(\infty)}) \rightarrow \pi(\boldsymbol{\theta})$  irrespective of the initial state.

*Irreducible:* the Markov chain can reach any state from any other state in a finite number of steps.

*Aperiodic:* the Markov chain does not get trapped in cycles.

*Detailed balance (reversible):* a Markov chain satisfies detailed balance if  $\pi(\boldsymbol{\theta}^{(i)})T(\boldsymbol{\theta}^{(i-1)} | \boldsymbol{\theta}^{(i)}) = \pi(\boldsymbol{\theta}^{(i-1)})T(\boldsymbol{\theta}^{(i)} | \boldsymbol{\theta}^{(i-1)})$ . This is a sufficient condition for an invariant distribution to exist.

### 3.1 Metropolis-Hastings Algorithm

Define the *transition kernel* as

$$\underbrace{P(x, dy)}_{\text{prob. of transition from } x \text{ to } dy} = \underbrace{p(x, y)dy}_{\text{prob. of moving to } dy \text{ around } y} + \underbrace{r(x)\delta_x(dy)}_{\text{prob. of staying at } x}$$

If  $p(x, x) = 0$  and  $\int_{\mathbb{R}^d} P(x, dy) = 1$ , then  $r(x) = 1 - \int_{\mathbb{R}^d} p(x, y)dy$  is the probability that the chain remains at  $x$ .

If  $p(x, y)$  satisfies detailed balance  $\pi(x)p(x, y) = \pi(y)p(y, x)$ , then  $\pi(\cdot)$  is the invariant density of the transition kernel  $P(x, \cdot)$ . In this case the transition kernel is reversible, i.e.  $P(x, dy)\pi(dx) = P(y, dx)\pi(dy)$ .

*\*Proof:*

$$\begin{aligned} \int_{\mathbb{R}^d} P(x, A)\pi(x)dx &= \int_{\mathbb{R}^d} \left[ \int_A p(x, y)dy \right] \pi(x)dx + \int_{\mathbb{R}^d} r(x)\delta_x(A)\pi(x)dx \\ &= \int_A \left[ \int_{\mathbb{R}^d} \pi(x)p(x, y)dx \right] dy + \int_A r(x)\pi(x)dx \\ &= \int_A \left[ \int_{\mathbb{R}^d} \pi(y)p(y, x)dx \right] dy + \int_A r(x)\pi(x)dx \\ &= \int_A [1 - r(y)]\pi(y)dy + \int_A r(y)\pi(y)dy \\ &= \int_A \pi(y)dy = \pi^*(A) \end{aligned}$$

To achieve this, the *proposal density*  $q(x, y)$  is balanced with an *acceptance probability* that satisfies  $\pi(x)q(x, y)\alpha(x, y) = \pi(y)q(y, x)\alpha(y, x)$ . Set  $\alpha(y, x) = 1$  then  $\alpha(x, y) = \frac{\pi(y)q(y, x)}{\pi(x)q(x, y)}$  if  $\pi(x)q(x, y) \geq \pi(y)q(y, x)$ ; otherwise  $\alpha(x, y) = 1$ . This gives  $p_{\text{MH}}(x, y) = q(x, y)\alpha(x, y)$  for  $x \neq y$  with the acceptance probability

$$\alpha(x, y) = \min \left( 1, \frac{\pi(y)q(y, x)}{\pi(x)q(x, y)} \right)$$

The transition kernel is

$$P_{\text{MH}}(x, dy) = q(x, y)\alpha(x, y)dy + \left(1 - \int_{\mathbb{R}^d} q(x, y)\alpha(x, y)dy\right) \delta_x(dy)$$

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**Algorithm 1** Metropolis-Hastings Algorithm

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Initialize  $x^{(1)}$ 
for  $j = 1 \rightarrow N$  do
    Simulate  $y \sim q(x^{(j)}, \cdot)$  ▷ Propose a candidate from proposal distribution
    Simulate  $u \sim U(0, 1)$  ▷ Generate a random number for acceptance decision
    if  $u \leq \alpha(x^{(j)}, y)$  then ▷ Acceptance probability  $\alpha(x, y) = \min\left(1, \frac{\pi(y)q(y, x)}{\pi(x)q(x, y)}\right)$ 
        Set  $x^{(j+1)} = y$  ▷ Accept the proposal and move to the new state
    else
        Set  $x^{(j+1)} = x^{(j)}$  ▷ Reject the proposal and stay at the current state
    end if
end for
Return  $\{x^{(1)}, x^{(2)}, \dots, x^{(N)}\}$ 

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### 3.2 Gibbs Sampling

**Product transition kernels:** the product kernel  $P_1(x_1, dy_1|x_2)P_2(x_2, dy_2|y_1)$  has invariant density  $\pi(y_1, y_2)$  if:

- the conditional kernel  $P_1(x_1, dy_1|x_2)$  has invariant density  $\pi_{1|2}(\cdot|x_2)$  for fixed  $x_2$
- the conditional kernel  $P_2(x_2, dy_2|x_1)$  has invariant density  $\pi_{2|1}(\cdot|x_1)$  for fixed  $x_1$ .

*\*Proof:*

$$\begin{aligned}
 & \int \int P_1(x_1, dy_1|x_2)P_2(x_2, dy_2|y_1)\pi(x_1, x_2)dx_1dx_2 \\
 &= \int P_2(x_2, dy_2|y_1) \left[ \int P_1(x_1, dy_1|x_2)\pi_{1|2}(x_1|x_2)dx_1 \right] \pi_2(x_2)dx_2 \\
 &= \int P_2(x_2, dy_2|y_1)\pi_{1|2}^*(dy_1|x_2)\pi_2(x_2)dx_2 \\
 &= \int P_2(x_2, dy_2|y_1)\pi_{2|1}(x_2|y_1)\pi_1^*(dy_1)dx_2 \\
 &= \pi_1^*(dy_1) \int P_2(x_2, dy_2|y_1)\pi_{2|1}(x_2|y_1)dx_2 \\
 &= \pi_1^*(dy_1)\pi_{2|1}^*(dy_2|y_1) = \pi^*(dy_1, dy_2)
 \end{aligned}$$

**Gibbs sampler:** set kernels to  $P_1(x_1, dy_1|x_2) = \pi_{1|2}^*(dy_1|x_2)$  and  $P_2(x_2, dy_2|x_1) = \pi_{2|1}^*(dy_2|x_1)$ . Use exact conditional densities as the proposal densities, i.e. for transition of  $x_1$  with  $x_2$  fixed,  $q([x_1, x_2], [y_1, y_2]) = \pi(y_1|x_2)$  and  $q([y_1, y_2], [x_1, x_2]) = \pi(x_1|y_2)$ . As a result, the acceptance probability is  $\alpha = 1$  for all moves.

Limitation: small step sizes and poor mixing if the components are highly correlated.

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**Algorithm 2** Gibbs Sampler

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```
1: for  $i = 1, \dots, N$  do
2:   for  $j = 1, \dots, d$  do ▷ Update one dimension at a time
3:      $x_j^{(i)} \sim \pi(\cdot \mid x_1^{(i)}, \dots, x_{j-1}^{(i)}, x_{j+1}^{(i-1)}, \dots, x_d^{(i-1)})$  ▷ Sample from the conditionals
4:   end for
5: end for
6: return  $x^{(0)}, \dots, x^{(N)}$ 
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**Mixture transition kernels:** mixing transitional kernels that have the same invariant density, i.e.  $P(x, dy) = \sum_i w_i P_i(x, dy)$  with  $\sum_i w_i = 1$ .

Advantage: for multimodal distributions, the mixture kernel can have better mixing properties by combining local and global moves.

### 3.3 Data Augmentation

Augment a target density  $\pi(\theta)$  with  $\phi$  to form  $\pi(\theta, \phi)$ . Draw samples  $\theta^{(n)}, \phi^{(n)} \sim \pi(\theta, \phi)$ , then for each  $\theta^{(n)}$  it follows  $\int \pi(\theta^{(n)}, \phi) d\phi = \pi(\theta^{(n)})$ .

## 4 Vector, Metric and Banach Spaces

**Vector space  $V$ :** a collection of vectors  $x$  with axioms (group):

- $x + y = y + x$  (commutativity)
- $(x + y) + z = x + (y + z)$  (associativity)
- There exists a zero vector such that  $x + 0 = x$  for all  $x$
- For each  $x$ , there exists an additive inverse such that  $x + (-x) = 0$
- For  $\alpha \in \mathbb{F}$ ,  $\alpha x \in V$  for all  $x \in V$  (scalar multiplication is closed)
- $\alpha(x + y) = \alpha x + \alpha y$  (distributivity)
- $\alpha(\beta x) = (\alpha\beta)x$  (associativity of scalar multiplication)
- $0 \times x = 0, 1 \times x = x$

**Inner product space:** space with an inner product  $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{F}$  satisfying:

- $\langle x, y \rangle = \overline{\langle y, x \rangle}$
- $\langle \alpha x + \beta y, z \rangle = \alpha \langle x, z \rangle + \beta \langle y, z \rangle$  for  $\alpha, \beta \in \mathbb{R}$
- $\langle x, x \rangle \geq 0$  and  $\langle x, x \rangle = 0$  if and only if  $x = 0$
- For  $a, b \in \mathbb{C}^n$ ,  $\langle a, b \rangle = \sum_{i=1}^n a_i^* b_i$

Inner product of functions:  $\langle f, g \rangle = \int_a^b f(x)g(x)dx$  for  $x \in [a, b] \in \mathbb{R}$ .

**Vector norm:** the length of a vector, satisfying:

- Schwarz inequality:  $|\langle x, y \rangle| \leq \|x\| \|y\|$

- Triangle inequality:  $\|x + y\| \leq \|x\| + \|y\|$
- Parallelogram law:  $\|x + y\|^2 + \|x - y\|^2 = 2(\|x\|^2 + \|y\|^2)$

**Orthogonality:** two elements  $x$  and  $y$  are orthogonal if and only if  $\langle x, y \rangle = 0$ . e.g.  $\sin(x)$  and  $\cos(x)$  are orthogonal on  $[-\pi, \pi]$ .

**Orthonormal basis:** For a vector space  $V$ , a *basis* is a *finite* set of linearly independent, orthogonal unit vectors that can express any vector in  $V$  as  $x = \sum_{i=1}^n \alpha_i e_i$ .

**Cauchy sequence:** a sequence of vectors  $\{x_n\}$  is a Cauchy sequence if for any  $\epsilon > 0$ , there exists a natural number  $N$  such that for all  $m, n > N$ ,  $\|x_n - x_m\| < \epsilon$ .

If every Cauchy sequence in a vector space converges to a limit that is *also in the vector space*, then the vector space is *complete*.

**Metric vector space:** a vector space with a distance function  $\rho(x, y)$  that maps two vectors to a non-negative real number, satisfying:

- $\rho(x, y) = 0$  if and only if  $x = y$
- $\rho(x, y) = \rho(y, x)$  (symmetry)
- $\rho(x, z) \leq \rho(x, y) + \rho(y, z)$  (triangle inequality)

**Normed space:** a metric space with distance function  $\rho(x, y) = \|x - y\|$  induced by a non-negative norm  $\|\cdot\|$  for all  $x, y \in V$ .

**Banach space:** a *complete normed space*. All finite-dimensional normed spaces are Banach spaces.

The  $l_p$  space is a Banach space where  $\sum_{i=1}^{\infty} |x_i|^p < \infty$  for  $p \geq 1$ . The norm is the  $l^p$  norm  $\|x\|_p = (\sum_{i=1}^n |x_i|^p)^{1/p}$ .

The  $L_p$  space is a Banach space where  $\int_a^b |f(x)|^p dx < \infty$ , where  $dx$  is the Lebesgue measure. The norm is the  $L^p$  norm  $\|f\|_p = \left(\int_a^b |f(x)|^p dx\right)^{1/p}$ . The  $L^p$  norm can be used to measure the complexity of a machine learning model.

## 5 Hilbert Spaces

**Pre-Hilbert space:** an incomplete inner product space.

**Hilbert space:** a *complete inner product space*.

Any  $f$  in a Hilbert space can be expressed as  $f = \sum_{i=1}^{\infty} \langle f, e_i \rangle e_i$ . The infinite set of infinite-dimensional orthonormal vectors  $\{e_i\}$  can be a basis if and only if

$$\{e_i\} \text{ is complete} \iff \|f\|^2 = \sum_{i=1}^{\infty} |\langle f, e_i \rangle|^2 \quad (\text{Parseval's identity})$$

$*l^2$  and  $L^2$  spaces are both Hilbert spaces.

The norm of a Hilbert space is induced by the inner product  $\|f\| = \sqrt{\langle f, f \rangle}$ . The  $L^2$  norm  $\|f - g\|_2$  measures the closeness of functions in the mean but not necessarily



pointwise. When  $\|f - g\|_2 = 0$ , there is *equivalence almost everywhere* except for sets of measure zero. Hence, the  $L^2$  space is a space of *equivalence classes of functions*.

**Equivalence of  $l^2$  and  $L^2$  spaces:** a function  $f \in L^2$  can be represented with a sequence  $\{c_k = \langle f, \phi_k \rangle\} \in l^2$  by Fourier series

$$f = \sum_{k=1}^{\infty} c_k \phi_k$$

where  $\{\phi_k\}$  is an orthonormal basis of  $L^2$ .

$l^2$  and  $L^2$  are isomorphic, i.e. there is a one-to-one correspondence between the elements of  $l^2$  and  $L^2$ .  $l^2$  is a coordinate system for  $L^2$ .

**Reproducing kernel Hilbert space (RKHS):** a Hilbert space of functions where point evaluation is continuous i.e. functions close in norm are also close pointwise.

The *reproducing kernel function*  $K_x = K(x, \cdot)$  reproduces function evaluation by inner product  $f(x) = \langle f(\cdot), K_x(\cdot) \rangle$ . The kernel function is symmetric and positive definite, and uniquely defines the RKHS.

**Function approximation in RKHS:** Given data  $\{(x_i, y_i)\}_{i=1}^n$  where  $y_i = f(x_i)$ , the regularised empirical error

$$\frac{1}{N} \sum_{n=1}^N |y_n - f(x_n)|^2 + \lambda \|f\|^2$$

is minimised by approximating functions of the form

$$\hat{f}(\cdot) = \sum_{n=1}^N \alpha_n K(x_n, \cdot)$$

*\*Solution:*

$$\begin{aligned} \sum_{n=1}^N |y_n - \hat{f}(x_n)|^2 &= \sum_{n=1}^N \left| y_n - \sum_{m=1}^N \alpha_m K(x_m, x_n) \right|^2 = (K\alpha - y)^\top (K\alpha - y) \\ \|\hat{f}\|^2 = \langle \hat{f}, \hat{f} \rangle &= \sum_{n=1}^N \sum_{m=1}^N \alpha_n \alpha_m \langle K(x_n, \cdot), K(x_m, \cdot) \rangle = \sum_{n=1}^N \sum_{m=1}^N \alpha_n \alpha_m K(x_n, x_m) = \alpha^\top K \alpha \end{aligned}$$

The third equality follows from the reproducing property of the kernel  $f = \langle f, k \rangle$ .

Taking derivative with respect to  $\alpha$  gives:

$$\frac{\partial}{\partial \alpha} \left[ \frac{1}{N} (K\alpha - y)^\top (K\alpha - y) + \lambda \alpha^\top K \alpha \right] = \frac{2}{N} K(K\alpha - y) + 2\lambda K \alpha$$

Setting the derivative to zero

$$\begin{aligned} \frac{1}{N} K(K\alpha - y) + \lambda K \alpha &= 0 \\ (K + N\lambda I) \alpha &= y \\ \alpha &= (K + N\lambda I)^{-1} y \end{aligned}$$

The term  $\lambda \|f\|^2$  penalises the complexity of the function and prevents overfitting.

## 6 Sobolev Spaces

**Sobolev space:** a Sobolev space  $W_p^k(\Omega)$  is a Banach space where functions  $f \in L^p(\Omega)$  have *weak derivatives*  $D^s f \in L^p(\Omega)$  for  $0 \leq s \leq k$ . The norm is defined as

$$\|f\|_{W_p^k} = \left( \sum_{s=0}^k \int_{\Omega} |D^s f(x)|^p dx \right)^{1/p}$$

For  $p = 2$ , the Sobolev space  $W_2^k(\Omega)$  is a Hilbert space with inner product

$$\langle f, g \rangle_{W_2^k} = \sum_{s=0}^k \langle D^s f, D^s g \rangle = \sum_{s=0}^k \int_{\Omega} D^s f(x) D^s g(x) dx$$

The norm and the inner product in  $W_2^k(\Omega)$  encodes the smoothness of function.

**Weak derivatives:** for functions  $f : [0, 1] \rightarrow \mathbb{R}$ , a function  $h : [0, 1] \rightarrow \mathbb{R}$  is a weak derivative of  $f$  if for every differentiable function  $g : [0, 1] \rightarrow \mathbb{R}$  with  $g(0) = g(1) = 0$ ,

$$\int_0^1 f(x) g'(x) dx = - \int_0^1 h(x) g(x) dx$$

**Function approximation by polynomial expansion:** By Fourier expansion, functions from the set  $C_2[-\pi, \pi] = C[-\pi, \pi] \cap L^2[-\pi, \pi]$  can be represented as  $f(x) = \sum_{k=0}^{\infty} c_k \exp(ikx)$  with  $c_k = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) \exp(-ikx) dx$ . The  $L^2$  norm of  $f$  is  $\|f\|^2 = \sum_{k=0}^{\infty} c_k^2$ .

The approximating functions  $f_n(x) = \sum_{k=0}^n c_k \exp(ikx)$  are in the Sobolev space  $W_2^s$  with the norm  $\|f\|_{W_2^s}^2 = \|D^s f\|_{L^2}^2 = \sum_{k=0}^{\infty} k^{2s} c_k^2$ .

\*The approximation error is bounded as

$$\begin{aligned} \|f - f_n\|_{L^2}^2 &= \sum_{k=n+1}^{\infty} c_k^2 = \sum_{k=n+1}^{\infty} \frac{1}{k^{2s}} k^{2s} c_k^2 \\ &< \frac{1}{n^{2s}} \sum_{k=n+1}^{\infty} k^{2s} c_k^2 < \frac{1}{n^{2s}} \sum_{k=1}^{\infty} k^{2s} c_k^2 \\ &= \frac{1}{n^{2s}} \|f\|_{W_2^s}^2 \end{aligned}$$

As  $s$  increases, the approximation error decreases faster with increasing  $n$ .

For  $d$ -dimensional domains, the error bound is

$$\|f - f_n\|_{L^2}^2 < \frac{1}{dn^{\frac{2s}{d}}} \|f\|_{W_2^s}^2$$

which shows the curse of dimensionality – convergence rate decreases as  $d$  increases.

## 7 Gaussian Measures in Hilbert Space

**Lebesgue measure in  $D$  dimensions:**  $\mu = \prod_{d=1}^D \mu_d$ .

There is no Lebesgue measure in infinite dimensional Hilbert or Banach spaces.

\*Proof:

- A Hilbert space  $H$  has an associated basis  $\{e_i\}_{i=1}^{\infty}$ .
- Denote a ball of radius  $r$  with center at  $x$  as  $B(x, r)$ .
- For a ball  $B(0, 2)$ , place a ball of radius  $\frac{1}{2}$  at each basis vector  $e_i$ :  $\{B(e_i, \frac{1}{2})\}_{i=1}^{\infty}$ .
- The balls do not intersect with each other:  $B(e_i, \frac{1}{2}) \cap B(e_j, \frac{1}{2}) = \emptyset$  for  $i \neq j$ .
- And the union of all balls is a subset of  $B(0, 2)$ :  $\bigcup_{i=1}^{\infty} B(e_i, \frac{1}{2}) \subset B(0, 2)$ .
- Since the Lebesgue measure in infinite dimensions is *translation invariant*,  $\mu(B(e_i, \frac{1}{2})) = \mu(B(e_j, \frac{1}{2}))$  for all  $i, j$ .
- *Monotonicity and countable additivity* gives  $\sum_{i=1}^{\infty} \mu(B(e_i, \frac{1}{2})) \leq \mu(B(0, 2))$ .
- The Lebesgue measure  $\mu(B(0, 2)) < \infty$ , but  $\sum_{i=1}^{\infty} \mu(B(e_i, \frac{1}{2})) = \infty$ , which is a contradiction. Therefore the Lebesgue measure is not well-defined in infinite dimensional Hilbert spaces.

**Gaussian measure:** The standard Gaussian measure is

$$g(B) = \frac{1}{\sqrt{2\pi}} \int_B \exp\left(-\frac{x^2}{2}\right) dx$$

For a product measure  $\mu = \prod_{d=1}^{\infty} g_k$  on  $\mathbb{R}^{\infty}$  to be well defined,  $\exp\left(-\frac{1}{2} \sum_{k=1}^{\infty} x_k^2\right)$  is finite and non-zero. So  $\sum_{k=1}^{\infty} x_k^2 < \infty$ , which is the definition of  $l_2$ :

$$x \in l_2 = \{x \in \mathbb{R}^{\infty} : \sum_{k=1}^{\infty} x_k^2 < \infty\}$$

**Gaussian measure in infinite dimensions:** A Gaussian measure is a probability measure characterised by a mean function and a covariance operator.

Diagonal covariance operator:  $S_x = (a_n x_n)_{n=1}^{\infty}$  for  $x \in l^2$  subject to  $\sum_{k=1}^{\infty} a_k < \infty$ .

Full covariance operators: need to be *trace class*, i.e. the trace  $\text{Tr}(C)$  is finite

$$\text{Tr}(C) = \sum_{n=1}^{\infty} \langle e_n, C e_n \rangle < \infty$$

The inner product  $\langle e_n, C e_n \rangle$  is

$$\begin{aligned} \langle e_n, C e_n \rangle &= \mathbb{E}[(e_n, u)(u, e_n)] \\ &= \mathbb{E} \left[ \int_{\Omega} \int_{\Omega} e_n(x) [u(x)u(y)] e_n(y) dy dx \right] \\ &= \mathbb{E} \left[ \int_{\Omega} e_n(x) \left( \int_{\Omega} u(x)u(y) e_n(y) dy \right) dx \right] \\ &= \int_{\Omega} e_n(x) \left( \int_{\Omega} \mathbb{E}[u(x)u(y)] e_n(y) dy \right) dx \\ &= \int_{\Omega} e_n(x) \left( \int_{\Omega} c(x, y) e_n(y) dy \right) dx \end{aligned}$$

where  $c(x, y) = \mathbb{E}[u(x)u(y)]$  is the covariance function and  $C e_n(x) = \int_{\Omega} c(x, y) e_n(y) dy$ .

**Gaussian measures in Hilbert space:** In a Hilbert space, two Gaussian measures are either equivalent or singular.

*Equivalent (mutually absolutely continuous):* Two measures are equivalent if they both assign zero measure to the same sets.

*Singular:* Two measures are singular w.r.t each other if one measure assigns zero to a set and the other measure assigns zero measure to the set complement.

For two Gaussian measures  $\mathcal{N}(0, C)$  and  $\mathcal{N}(v, C)$  in  $l^2$ , two measures are equivalent if  $v \in \text{Image}(C^{1/2})$  i.e.  $v = C^{1/2}u$  for some  $u \in l^2$ , and singular otherwise. The subspace for  $v$  is the Cameron-Martin space (aka the reproducing kernel Hilbert space).

**Bayes' rule in Hilbert space:** The Radon-Nikodym derivative of the posterior probability measure with respect to the prior probability measure is given by

$$\frac{d\mu^y}{d\mu^0} \propto \underbrace{P(y|x)}_{\text{likelihood}}$$

In a infinite dimensional Hilbert space, the prior measure over functions is a Gaussian measure  $\mu_0 = \mathcal{N}(m, C)$  – formal definition of Gaussian Process prior.

## 8 MCMC in Infinite-Dimensional Spaces

**Metropolis-Hastings algorithm in infinite dimensions:** Define a proposal

$$v = \sqrt{1 - \beta^2}u + \beta\xi, \xi \sim \mathcal{N}(0, C)$$

such that  $u$  and  $v$  both follow  $\mathcal{N}(0, C)$ . Given a prior Gaussian measure  $\mu^0 = \mathcal{N}(0, C)$  and a likelihood  $P(y|v) = \exp(-\Phi(v))$ , the acceptance probability (replacing  $\pi(v)q(v, u)$  with  $\mu^y(dv)q(dv, u)$ , where  $\mu^y$  is the posterior measure) is

$$\alpha(u, v) = \min \left( 1, \frac{\mu^y(dv)q(v, du)}{\mu^y(du)q(u, dv)} \right) = \min \left( 1, \frac{P(y|v)}{P(y|u)} \right) = \min(1, \exp(\Phi(u) - \Phi(v)))$$

For  $v = u + \beta\xi, \xi \sim \mathcal{N}(0, C)$  the acceptance probability is  $\min(1, \exp(J(v) - J(u)))$ , where  $J(v) = -\Phi(v) - \frac{1}{2}|C^{-1/2}v|^2$ . The expectation of the norm  $\mathbb{E}[|C^{-1/2}v|^2]$  is infinite.

## 9 Langevin Dynamics

**Langevin stochastic differential equation (SDE):**

$$dX_t = -\nabla_x U(X_t)dt + \sqrt{2}dB_t = \nabla_x \log \pi(X_t)dt + \sqrt{2}dB_t$$

where  $U(\cdot) : \mathbb{R}^d \rightarrow \mathbb{R}$  is a potential function,  $(B_t)_{t \geq 0}$  is the Brownian motion and  $\pi(x) \propto \exp(-U(x))$  is the stationary distribution of the SDE.

The measure of  $X_t, \rho_t$ , satisfies the Fokker-Planck equation

$$\frac{\partial \rho_t}{\partial t} = \nabla \cdot (\rho_t \nabla \log \pi) + \Delta \rho_t$$

with asymptotic solution  $\rho_\infty = \pi$ .

**Log-concave:** A probability measure with density  $\pi$  is log-concave if

$$\pi(\lambda x + (1 - \lambda)y) \geq \pi(x)^\lambda \pi(y)^{1-\lambda} \text{ for } 0 < \lambda < 1$$

A probability measure with density  $\pi$  is m-strongly log-concave if

$$\pi(\lambda x + (1 - \lambda)y) \geq \pi(x)^\lambda \pi(y)^{1-\lambda} \exp\left(\frac{m}{2} \lambda(1 - \lambda) \|x - y\|^2\right) \text{ for } 0 < \lambda < 1$$

i.e. there are no flat regions and the maximum is unique. For one dimension

$$\nabla^2 \log \pi(x) \leq -m < 0 \quad \forall x$$

**Convergence of Langevin dynamics:** For an m-strongly log-concave  $\pi$ , the convergence rate to the stationary measure is exponentially fast

$$W_2(\rho_t, \pi) \leq e^{-mt} \left\{ \|x - x^*\| + \left(\frac{d}{m}\right)^{\frac{1}{2}} \right\}$$

for  $\pi_0 = \delta_x$  and  $x^* = \arg \max_x \pi(x)$ .

**Metropolis-adjusted Langevin algorithm (MALA):** Given a state of the Markov chain  $X_k$ , propose a new state  $\bar{X}_{k+1}$  by first-order Euler discretisation with step size  $\gamma$ :

$$\bar{X}_{k+1} = X_k + \gamma \nabla \log \pi(X_k) + \sqrt{2\gamma} W_k, W_k \sim \mathcal{N}(0, I)$$

To correct the discretisation bias, accept the proposed move with probability

$$\alpha(X_k, \bar{X}_{k+1}) = \min \left( 1, \frac{\pi(\bar{X}_{k+1}) q(X_k | \bar{X}_{k+1})}{\pi(X_k) q(\bar{X}_{k+1} | X_k)} \right)$$

where  $q(\bar{X}_{k+1} | X_k) \propto \exp \left( -\frac{1}{4\gamma} \|\bar{X}_{k+1} - X_k - \gamma \nabla \log \pi(X_k)\|^2 \right)$

Limitation: the acceptance probability is exponentially low for high dimension, leading to poor mixing.

**Unadjusted Langevin algorithm (ULA):** Run the first-order Euler discretisation without the Metropolis-Hastings correction, i.e.  $X_{k+1} = X_k + \gamma \nabla \log \pi(X_k) + \sqrt{2\gamma} W_k$ . The stationary distribution is a biased distribution  $\pi_\gamma$ , with asymptotic bias of order  $\mathcal{O}(\gamma^{\frac{1}{2}})$ .

**Langevin schemes for Bayesian inference:** For a target distribution  $\pi(x) \propto p(x) \prod_{i=1}^n p(y_i | x)$ , where  $p(x)$  is the prior and  $p(y_i | x)$  is the likelihood for data vector  $y_i$ , samples can be generated from

$$X_{k+1} = X_k + \gamma \left( \nabla \log p(X_k) + \sum_{i=1}^n \nabla \log p(y_i | X_k) \right) + \sqrt{2\gamma} W_k$$

For MALA, the complexity is  $\mathcal{O}(n)$  for both the gradient and the acceptance probability (can be avoided in ULA) computation.

**Stochastic gradient Langevin dynamics (SGLD):** An unbiased estimate of the gradient can be computed more efficiently by subsampling the data. At each iteration  $k$ , choose a random index set  $\mathcal{I}_k \subset \{1, \dots, n\}$  and approximate the gradient

$$\nabla \log \widehat{\pi}(X_k) = \nabla \log p(X_k) + \frac{n}{|\mathcal{I}_k|} \sum_{i \in \mathcal{I}_k} \nabla \log p(y_i | X_k)$$

The complexity is  $\mathcal{O}(|\mathcal{I}_k|)$ , and convergence rate is similar to ULA.

## 10 Energy-Based and Score-Based Models

**Generative models:** approximate the data distribution  $p_{\text{data}}$  with a model distribution  $p_{\theta}$ , by learning parameters  $\theta$  from data  $\mathcal{D} = \{x_i\}_{i=1}^N$  drawn from  $p_{\text{data}}$ .

**Energy-based models:** The probability of a state  $x$  is defined as

$$p(x) = \frac{1}{Z_{\beta}} \exp(-\beta E(x))$$

where  $E(x)$  is the energy function,  $\beta$  is the inverse temperature and  $Z_{\beta} = \int \exp(-\beta E(x)) dx$  is the partition function. EBMs parameterise the energy function, giving  $p_{\theta}(x) \propto \exp(-E_{\theta}(x))$ .

The EBMs are trained via maximum likelihood or contrastive divergence.

*Maximum likelihood:* maximise the log-likelihood  $\mathcal{L}(\theta) = \mathbb{E}_{x \sim p_{\text{data}}} [\log p_{\theta}(x)]$  via gradient methods, with gradient:

$$\nabla_{\theta} \mathcal{L}(\theta) = -\mathbb{E}_{x \sim p_{\text{data}}} [-\nabla_{\theta} E_{\theta}(x)] + \underbrace{\mathbb{E}_{x \sim p_{\theta}} [-\nabla_{\theta} E_{\theta}(x)]}_{\nabla_{\theta} \log Z_{\theta}}$$

Limitation: the second term requires sampling from the model distribution  $p_{\theta}$ , which is computationally expensive.

*Contrastive divergence:*

1. For each  $x_i$ , initialise a MCMC chain with target distribution  $p_{\theta}$  at  $x_i^{(0)}$ .
2. Run the MCMC chain for  $k$  steps to obtain  $x_i^{(k)}$ .
3. Compute the gradient estimate

$$\hat{\mathcal{L}} = -\frac{1}{N} \sum_{i=1}^N \left( -\nabla_{\theta} E_{\theta}(x_i^{(0)}) + \nabla_{\theta} E_{\theta}(x_i^{(k)}) \right)$$

4. Update parameters  $\theta \leftarrow \theta + \eta \hat{\mathcal{L}}$ .

Limitations: bias in the gradient estimate and MCMC can be very slow.

**Score-based models:** Learn to match the score function  $s_{\theta}(x) = \nabla_x \log p_{\theta}(x) = -\nabla_x E_{\theta}(x)$ , bypassing the partition function. Trained by score matching, which minimises the Fisher divergence between  $p_{\text{data}}$  and  $p_{\theta}$ :

$$J(\theta) = \frac{1}{2} \mathbb{E}_{x \sim p_{\text{data}}} \left[ \left\| s_{\theta}(x) - \underbrace{\nabla_x \log p_{\text{data}}(x)}_{\text{intractable}} \right\|^2 \right]$$

which can be rewritten as

$$J_{\text{SM}}(\theta) = \mathbb{E}_{x \sim p_{\text{data}}} \left[ \underbrace{\text{Tr}(\nabla_x s_{\theta}(x))}_{\text{expensive for high-dim.}} + \frac{1}{2} \|s_{\theta}(x)\|^2 \right]$$

*Sliced score matching*: approximate the trace of the Jacobian by projecting the score function onto random directions  $v \sim \mathcal{N}(0, I)$

$$J_{\text{SSM}}(\theta) = \mathbb{E}_{x \sim p_{\text{data}}, v \sim \mathcal{N}(0, I)} \left[ v^{\top} \nabla_x s_{\theta}(x) v + \frac{1}{2} \|s_{\theta}(x)\|^2 \right]$$

In practice, one sample of  $v$  is used for each data point:

$$\hat{J}_{\text{SSM}}(\theta) = \frac{1}{N} \sum_{i=1}^N \left[ v_i^{\top} \nabla_x s_{\theta}(x_i) v_i + \frac{1}{2} \|s_{\theta}(x_i)\|^2 \right]$$

**Sampling from generative models:** use Langevin dynamics

$$dx_t = s_{\theta}(x) dt + \sqrt{2} dB_t$$

which have stationary distribution  $p_{\theta}$ . In practice, discretise the SDE with step size  $\gamma$ :

$$x^{(k+1)} = x^{(k)} + \gamma s_{\theta}(x^{(k)}) + \sqrt{2\gamma} z^{(k)}, \quad z^{(k)} \sim \mathcal{N}(0, I)$$

**Parallel tempering (replica exchange):** For  $\pi(x) = \exp(-U(x))$ , define the tempered (smoothed) distribution  $\pi_{\beta}(x) = \exp(-\beta U(x))$  with inverse temperature  $\beta$ . Run MCMC chains on multiple tempered distributions  $\{\pi_{\beta_i}\}$  in parallel, and exchange samples between  $i$ -th and  $j$ -th chains with acceptance probability:

$$\alpha = \min \left( 1, \frac{\pi_{\beta_i}(x_t^j) \pi_{\beta_j}(x_t^i)}{\pi_{\beta_i}(x_t^i) \pi_{\beta_j}(x_t^j)} \right)$$